# Business Statistics

## Value-based measures of dispersion



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

# Measures of Dispersion Overview

- The dispersion of a variable has two main characteristics:
  - Spread - the distance observations are from the center
  - Shape - how observations are distributed around the center
- Value-based dispersions
  - Mean Absolute Deviation (MAD)
  - Mean Absolute Percentage Deviation (MAPD)
  - Variance
  - Standard Deviation
  - Coefficient of Variation
  - Skew
  - Kurtosis

# Dispersion of a variable

## Value-based measures of dispersion

# Dispersion of a Variable
# A variable takes on multiple values

- As observations have different values, measures of **dispersion** (spread) describe how much those values differ from the center
- Greater dispersion indicates greater variability among the observations
- The observations' dispersion affects:
  - How likely future observations are to be close to the center
  - The variability of calculations based on those observations
- Different measures give different information, so we consider multiple measures to get the whole picture

# Value-based measures of dispersion
# Calculating the deviation of all observations

- Recall the mean is the value-based measure of "center" for a variable
- Value-based measures depend on how each observation deviates from the mean
- An observation's "deviation" is the observation's value minus the mean
  - For values greater than the mean this results in a positive number
  - For values less than the mean this results in a negative number

$$x_i - \overline{x}$$

- Summing all the deviations **MUST** always = 0
  - The mean was calculated to balance out all the observations
  - Therefore, summing all the deviations cancel out the positive and negative values
- Therefore, to measure the total amount of deviation (dispersion) we must prevent this "cancelling out" effect

# Value-based measures of dispersion
# Different measures high-level

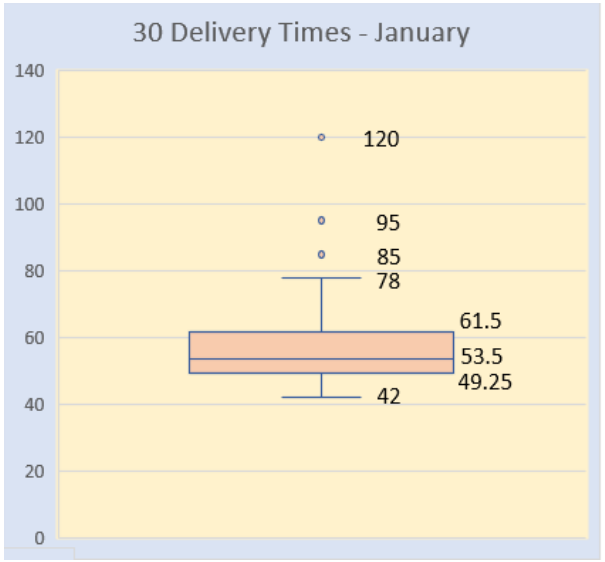
- The measures we will consider are:
  - Mean Absolute Deviation (MAD)
  - Mean Absolute Percentage Deviation (MAPD)
  - Variance
  - Standard Deviation
  - Coefficient of Variation
  - Skew
  - Kurtosis
- MAD and variance use different approaches to prevent positive and negative deviations from canceling each other out

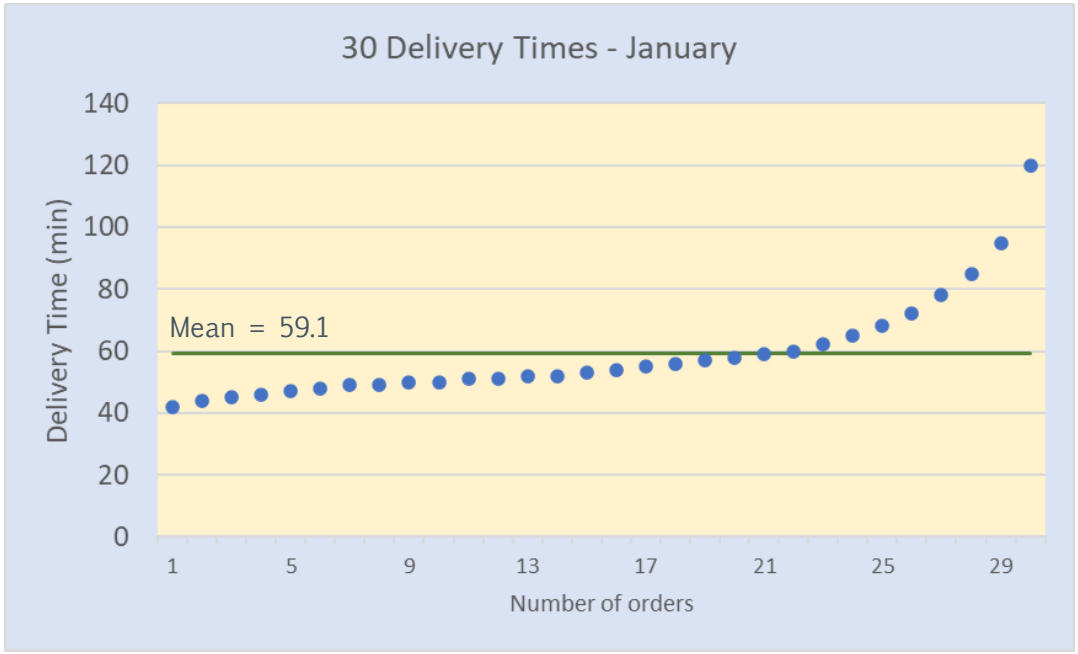
# Value-based measures of dispersion
# Data set and ordered measures

- An online retailer tracks the delivery time (in minutes) for 30 recent customer orders shipped from the same warehouse in January.
- Management wants to understand typical delivery time, variability, consistency, and whether extreme delays exist.

| Delivery Time |
|---|
| 46 |
| 65 |
| 48 |
| 55 |
| 120 |
| ... |
| 50 |
| 62 |
| 49 |
| 42 |
| 51 |

| 5 Number Summary January | | | | |
|---|---|---|---|---|
| Min | Q1 | Q2 | Q3 | Max |
| 42 | 49.25 | 53.5 | 61.5 | 120 |



# Dispersion of a variable

## Variance



# Value-based measures of dispersion
# Variance

- Variance: How do observations "vary" from the mean?
- Preventing the "cancelling out" of summing positive and negative deviations:
  - For MAD the absolute value was taken for all the deviations before summing them
  - For variance, the deviations are squared before summing them
  - A squared number is always positive
- Impact:
  - For MAD, each observation contributes **proportionally** to the sum of absolute deviations
  - For **variance**, each observation contributes **the square of its deviation** to the sum
  - Therefore, larger deviations have greater impact on the variance
- Squared deviations have squared units which are difficult to interpret

# Value-based measures of dispersion
# Variance: $s^2$ for the sample and $\sigma^2$ for the population

- Take the deviation of each observation $(x_i - \overline{x})$ and square it:
  - For sample deviations this is $(x_i - \overline{x})^2$
- Sum the squared deviations :
  - Sum of these squared deviations is $\sum_i (x_i - \overline{x})^2$
- Divide this sum of the squared deviations:
  - For the **sample** divide by $(n - 1)$
  - For the **population** divide by $N$

Sample variance: $s^2 = \dfrac{\sum_i (x_i - \overline{x})^2}{n - 1}$

Population variance: $\sigma^2 = \dfrac{\sum_i (x_i - \mu)^2}{N}$

Excel:

Sample: =VAR.S( <data range> )

Population: =VAR.P( <data range> )

# Value-based measures of dispersion
# Variance: delivery times in minutes (mins)

- Calculate the deviation of each observation $x_i$ from the sample mean $\overline{x}$
  - For the delivery times sample this is $(x_i\ mins - 59.1\ mins)$
- Square all the deviations and then sum them:
  - For the delivery times this is $\sum_i (x_i\ mins - 59.1\ mins)^2$
- Divide this sum of squared deviations by $n - 1$
  - For the delivery times sample divide by $(30 - 1) = 29$

Sample variance: $s^2 = \dfrac{\sum_i (x_i - \overline{x})^2}{n-1} = \dfrac{\sum_i (x_i\ mins\ - 59.1\ mins)^2}{29}$

$$s^2 = 281.82\ mins^2$$

# Dispersion of a variable

## Standard Deviation

# Value-based measures of dispersion
# Standard Deviation: abbreviated SD or Std. Dev.

- The major issue with variance is the interpretation of squared units.
  - What does $s^2 = 281.8\ mins^2$ mean?
- Standard deviation (SD) is based on the variance but is in the original units
- Original units enable comparing spread to the mean
  - A relatively low SD occurs when observations are closer to the mean
  - A relatively high SD occurs when observations are spread out further from the mean
- Being based on the variance, the larger an observation's deviation, the greater its impact on dispersion as measured by the SD

# Value-based measures of dispersion
# Standard Deviation calculation

- The standard deviation (SD) is the square root of the variance
- While the SD is in the original units, it is still not as intuitive as the mean
- The majority of observations lie within the range of the **mean ± SD**

Sample SD: $s = \sqrt{s^2}$

Population SD: $\sigma = \sqrt{\sigma^2}$

Excel:

Sample: = STDEV.S ( <data range> )

Population: = STDEV.P( <data range> )

# Value-based measures of dispersion
# Standard Deviation: delivery times in minutes (mins)

- The variance must first be calculated as the sum of squared deviations from the mean all divided by:
  - $n-1$ if calculated for the sample
  - $N$ if calculated for the population

$$s^2 = \frac{\sum_i (x_i - \overline{x})^2}{n-1} = \frac{\sum_i (x_i\ mins\ - 59.1\ mins)^2}{29} = 281.82\ mins^2$$

- The standard deviation is then the square root of that variance

$$s = \sqrt{s^2} = \sqrt{281.82\ mins^2} = \boxed{16.8\ mins}$$

# Dispersion of a variable

## Coefficient of Variation

# Value-based measures of dispersion
# How dispersed is the data from the mean?

- Standard deviation provides an absolute measure of deviation – $16.79\ mins$
- Again, this does not tell us how that SD compares to the mean
- Comparing multiple variables' variability requires an adjustment for the scale - a **relative** measure of the spread of a variable's values around the mean
- The Coefficient of Variation (CV) equals the SD divided by the mean, giving a unit-free, relative measure

Sample CV: $CV = s/\overline{x}$

Population CV: $CV = \sigma/\mu$

For the delivery times example:

$$CV = {}^{s}/_{\overline{x}} = {}^{16.8}/_{59.1} = 0.28 = \boxed{28\%}$$